

**KACHUG, Russian Federation**WMO: **306220**Lat: **53.967N**Lon: **105.900E**Elev: **521**StdP: **95.22**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |     |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|-----|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%  |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS  | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l) | (m)   | (n)                   | (o)  | (p)   |
| 1             | -43.0      | -40.4 | -45.2                         | 0.0 | -42.5 | -42.9 | 0.1 | -40.2 | 5.9                   | -14.7 | 4.9 | -15.4 | 0.1                   | 160  | 0.638 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 7             | 12.9                      | 29.5            | 18.7 | 27.6 | 17.8 | 25.7 | 17.1 | 20.3                | 26.5 | 19.2 | 24.9 | 18.2 | 23.7 | 2.0                     | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.2                            | 14.0 | 23.0 | 17.2 | 13.1 | 21.9 | 16.1 | 12.2 | 21.0 | 61.2          | 26.3 | 57.2 | 25.1 | 54.0 | 23.9 | 27.4              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |            |            | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |             |              |             |    |    |
|-------------------|------------|------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|----|----|
|                   |            |            | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |             | n=50 years   |             |    |    |
| 1%                | 2.5%       | 5%         | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max         | Min          | Max         | DB | WB |
| (a)               | (b)        | (c)        | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          | (p)         |    |    |
| (4) <b>5.8</b>    | <b>4.8</b> | <b>4.1</b> | <b>-44.0</b>               | <b>33.1</b> | <b>3.5</b>         | <b>1.7</b> | <b>-46.5</b>                                       | <b>34.3</b> | <b>-48.6</b> | <b>35.4</b> | <b>-50.6</b> | <b>36.3</b> | <b>-53.2</b> | <b>37.6</b> |    |    |
| (5)               |            |            | <b>-43.6</b>               | <b>22.7</b> | <b>3.1</b>         | <b>1.9</b> | <b>-45.9</b>                                       | <b>24.0</b> | <b>-47.7</b> | <b>25.1</b> | <b>-49.4</b> | <b>26.1</b> | <b>-51.7</b> | <b>27.5</b> |    |    |

**Monthly Climatic Design Conditions**

|                                                                               |             | Annual       | Jan          | Feb          | Mar          | Apr          | May          | Jun          | Jul          | Aug          | Sep          | Oct          | Nov          | Dec          |
|-------------------------------------------------------------------------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                                                                               |             | (d)          | (e)          | (f)          | (g)          | (h)          | (i)          | (j)          | (k)          | (l)          | (m)          | (n)          | (o)          | (p)          |
| (6) <b>Temperatures, Degree-Days and Degree-Hours</b>                         | DBAvg       | <b>-3.0</b>  | <b>-27.7</b> | <b>-23.1</b> | <b>-11.4</b> | <b>1.1</b>   | <b>8.3</b>   | <b>15.6</b>  | <b>18.4</b>  | <b>15.4</b>  | <b>7.6</b>   | <b>-1.5</b>  | <b>-15.1</b> | <b>-25.0</b> |
|                                                                               | DBStd       | <b>17.13</b> | <b>7.69</b>  | <b>7.51</b>  | <b>7.30</b>  | <b>4.48</b>  | <b>4.28</b>  | <b>3.74</b>  | <b>2.77</b>  | <b>3.37</b>  | <b>4.04</b>  | <b>5.15</b>  | <b>8.13</b>  | <b>7.97</b>  |
|                                                                               | HDD10.0     | <b>5408</b>  | <b>1168</b>  | <b>927</b>   | <b>662</b>   | <b>267</b>   | <b>85</b>    | <b>4</b>     | <b>0</b>     | <b>4</b>     | <b>95</b>    | <b>357</b>   | <b>752</b>   | <b>1087</b>  |
|                                                                               | HDD18.3     | <b>7852</b>  | <b>1426</b>  | <b>1160</b>  | <b>921</b>   | <b>517</b>   | <b>312</b>   | <b>97</b>    | <b>34</b>    | <b>102</b>   | <b>322</b>   | <b>615</b>   | <b>1002</b>  | <b>1345</b>  |
|                                                                               | CDD10.0     | <b>656</b>   | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>1</b>     | <b>32</b>    | <b>171</b>   | <b>260</b>   | <b>169</b>   | <b>23</b>    | <b>1</b>     | <b>0</b>     | <b>0</b>     |
|                                                                               | CDD18.3     | <b>59</b>    | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>1</b>     | <b>14</b>    | <b>36</b>    | <b>9</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     |
|                                                                               | CDH23.3     | <b>1007</b>  | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>49</b>    | <b>307</b>   | <b>464</b>   | <b>174</b>   | <b>12</b>    | <b>0</b>     | <b>0</b>     | <b>0</b>     |
|                                                                               | CDH26.7     | <b>260</b>   | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>9</b>     | <b>80</b>    | <b>133</b>   | <b>37</b>    | <b>1</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     |
|                                                                               | WSAvg       | <b>1.3</b>   | <b>0.5</b>   | <b>0.6</b>   | <b>1.2</b>   | <b>2.0</b>   | <b>2.2</b>   | <b>1.8</b>   | <b>1.5</b>   | <b>1.4</b>   | <b>1.5</b>   | <b>1.3</b>   | <b>0.8</b>   | <b>0.5</b>   |
| (15) <b>Precipitation</b>                                                     | PrecAvg     | <b>311</b>   | <b>7</b>     | <b>5</b>     | <b>4</b>     | <b>10</b>    | <b>23</b>    | <b>49</b>    | <b>66</b>    | <b>76</b>    | <b>37</b>    | <b>12</b>    | <b>11</b>    | <b>10</b>    |
|                                                                               | PrecMax     | <b>444</b>   | <b>15</b>    | <b>13</b>    | <b>15</b>    | <b>23</b>    | <b>49</b>    | <b>115</b>   | <b>174</b>   | <b>158</b>   | <b>86</b>    | <b>32</b>    | <b>20</b>    | <b>25</b>    |
|                                                                               | PrecMin     | <b>221</b>   | <b>1</b>     | <b>1</b>     | <b>0</b>     | <b>1</b>     | <b>5</b>     | <b>12</b>    | <b>19</b>    | <b>27</b>    | <b>10</b>    | <b>0</b>     | <b>2</b>     | <b>4</b>     |
|                                                                               | PrecStd     | <b>57</b>    | <b>3</b>     | <b>3</b>     | <b>3</b>     | <b>6</b>     | <b>12</b>    | <b>25</b>    | <b>37</b>    | <b>33</b>    | <b>20</b>    | <b>8</b>     | <b>4</b>     | <b>5</b>     |
|                                                                               |             |              |              |              |              |              |              |              |              |              |              |              |              |              |
| (19) <b>Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures</b> | 0.4%        | DB           | <b>-5.7</b>  | <b>-1.8</b>  | <b>7.6</b>   | <b>19.5</b>  | <b>27.9</b>  | <b>31.6</b>  | <b>32.2</b>  | <b>30.2</b>  | <b>24.9</b>  | <b>15.6</b>  | <b>3.6</b>   | <b>-4.5</b>  |
|                                                                               |             | MCWB         | <b>-6.5</b>  | <b>-3.5</b>  | <b>2.6</b>   | <b>9.2</b>   | <b>15.0</b>  | <b>18.1</b>  | <b>20.6</b>  | <b>19.0</b>  | <b>15.2</b>  | <b>9.0</b>   | <b>0.6</b>   | <b>-5.7</b>  |
|                                                                               | 2%          | DB           | <b>-9.8</b>  | <b>-5.6</b>  | <b>4.5</b>   | <b>15.5</b>  | <b>24.4</b>  | <b>28.9</b>  | <b>29.9</b>  | <b>27.3</b>  | <b>21.2</b>  | <b>11.3</b>  | <b>0.5</b>   | <b>-6.9</b>  |
|                                                                               |             | MCWB         | <b>-10.5</b> | <b>-6.9</b>  | <b>0.5</b>   | <b>7.2</b>   | <b>13.2</b>  | <b>17.5</b>  | <b>19.7</b>  | <b>17.9</b>  | <b>13.5</b>  | <b>6.1</b>   | <b>-1.2</b>  | <b>-7.6</b>  |
|                                                                               | 5%          | DB           | <b>-13.3</b> | <b>-8.8</b>  | <b>2.0</b>   | <b>12.3</b>  | <b>21.6</b>  | <b>26.8</b>  | <b>28.0</b>  | <b>25.2</b>  | <b>17.9</b>  | <b>8.4</b>   | <b>-1.5</b>  | <b>-10.2</b> |
|                                                                               |             | MCWB         | <b>-13.9</b> | <b>-9.9</b>  | <b>-1.0</b>  | <b>5.4</b>   | <b>11.5</b>  | <b>16.7</b>  | <b>18.8</b>  | <b>17.2</b>  | <b>11.8</b>  | <b>4.6</b>   | <b>-2.8</b>  | <b>-10.8</b> |
|                                                                               | 10%         | DB           | <b>-16.5</b> | <b>-11.9</b> | <b>-0.2</b>  | <b>9.5</b>   | <b>18.4</b>  | <b>24.8</b>  | <b>25.9</b>  | <b>23.1</b>  | <b>15.1</b>  | <b>5.9</b>   | <b>-3.8</b>  | <b>-13.2</b> |
|                                                                               |             | MCWB         | <b>-16.9</b> | <b>-12.6</b> | <b>-2.4</b>  | <b>3.9</b>   | <b>9.9</b>   | <b>16.0</b>  | <b>18.1</b>  | <b>16.4</b>  | <b>10.8</b>  | <b>2.9</b>   | <b>-4.8</b>  | <b>-13.7</b> |
|                                                                               | 0.4%        | WB           | <b>-6.4</b>  | <b>-3.4</b>  | <b>3.0</b>   | <b>9.8</b>   | <b>15.9</b>  | <b>20.6</b>  | <b>22.7</b>  | <b>20.9</b>  | <b>16.2</b>  | <b>9.3</b>   | <b>0.9</b>   | <b>-5.5</b>  |
|                                                                               |             | MCDB         | <b>-5.5</b>  | <b>-1.6</b>  | <b>7.0</b>   | <b>17.4</b>  | <b>25.9</b>  | <b>27.5</b>  | <b>29.5</b>  | <b>26.9</b>  | <b>22.6</b>  | <b>14.6</b>  | <b>3.3</b>   | <b>-4.7</b>  |
| (27) <b>Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures</b> | 2%          | WB           | <b>-10.5</b> | <b>-6.9</b>  | <b>0.7</b>   | <b>7.7</b>   | <b>13.7</b>  | <b>19.0</b>  | <b>21.0</b>  | <b>19.1</b>  | <b>14.3</b>  | <b>6.8</b>   | <b>-1.1</b>  | <b>-7.6</b>  |
|                                                                               |             | MCDB         | <b>-9.7</b>  | <b>-5.5</b>  | <b>4.0</b>   | <b>14.6</b>  | <b>23.0</b>  | <b>25.6</b>  | <b>27.6</b>  | <b>25.2</b>  | <b>20.0</b>  | <b>10.6</b>  | <b>0.3</b>   | <b>-6.9</b>  |
|                                                                               | 5%          | WB           | <b>-13.9</b> | <b>-10.0</b> | <b>-0.8</b>  | <b>5.9</b>   | <b>12.0</b>  | <b>17.9</b>  | <b>19.9</b>  | <b>18.0</b>  | <b>12.6</b>  | <b>4.9</b>   | <b>-2.8</b>  | <b>-10.7</b> |
|                                                                               |             | MCDB         | <b>-13.4</b> | <b>-8.9</b>  | <b>1.8</b>   | <b>11.6</b>  | <b>20.3</b>  | <b>24.8</b>  | <b>25.6</b>  | <b>23.5</b>  | <b>17.2</b>  | <b>8.0</b>   | <b>-1.5</b>  | <b>-10.2</b> |
|                                                                               | 10%         | WB           | <b>-16.9</b> | <b>-12.7</b> | <b>-2.5</b>  | <b>4.1</b>   | <b>10.4</b>  | <b>16.8</b>  | <b>18.9</b>  | <b>16.9</b>  | <b>11.0</b>  | <b>3.1</b>   | <b>-4.8</b>  | <b>-13.7</b> |
|                                                                               |             | MCDB         | <b>-16.5</b> | <b>-11.9</b> | <b>-0.2</b>  | <b>9.2</b>   | <b>17.4</b>  | <b>23.2</b>  | <b>24.2</b>  | <b>22.1</b>  | <b>14.7</b>  | <b>5.8</b>   | <b>-3.8</b>  | <b>-13.3</b> |
|                                                                               | 5% DB       | MDBR         | <b>9.0</b>   | <b>12.2</b>  | <b>14.8</b>  | <b>12.7</b>  | <b>14.8</b>  | <b>14.5</b>  | <b>12.9</b>  | <b>13.1</b>  | <b>12.6</b>  | <b>11.1</b>  | <b>9.8</b>   | <b>8.6</b>   |
|                                                                               |             | MCDBR        | <b>11.0</b>  | <b>14.1</b>  | <b>14.7</b>  | <b>18.2</b>  | <b>21.6</b>  | <b>19.1</b>  | <b>17.8</b>  | <b>17.7</b>  | <b>18.7</b>  | <b>16.0</b>  | <b>10.7</b>  | <b>10.5</b>  |
|                                                                               |             | MCWBR        | <b>10.4</b>  | <b>12.7</b>  | <b>11.1</b>  | <b>10.7</b>  | <b>11.4</b>  | <b>8.8</b>   | <b>8.3</b>   | <b>8.9</b>   | <b>10.7</b>  | <b>10.8</b>  | <b>9.1</b>   | <b>10.0</b>  |
|                                                                               |             | MCWBR        | <b>11.0</b>  | <b>14.1</b>  | <b>13.9</b>  | <b>17.1</b>  | <b>19.8</b>  | <b>15.9</b>  | <b>14.3</b>  | <b>15.3</b>  | <b>16.4</b>  | <b>15.4</b>  | <b>10.6</b>  | <b>10.5</b>  |
| (35) <b>Mean Daily Temperature Range</b>                                      | 5% WB       | MCWBR        | <b>10.4</b>  | <b>12.7</b>  | <b>10.6</b>  | <b>10.6</b>  | <b>10.8</b>  | <b>8.2</b>   | <b>7.5</b>   | <b>8.4</b>   | <b>9.7</b>   | <b>10.8</b>  | <b>9.1</b>   | <b>10.0</b>  |
|                                                                               |             |              |              |              |              |              |              |              |              |              |              |              |              |              |
|                                                                               | taub        |              | <b>0.273</b> | <b>0.302</b> | <b>0.329</b> | <b>0.370</b> | <b>0.414</b> | <b>0.458</b> | <b>0.447</b> | <b>0.391</b> | <b>0.343</b> | <b>0.318</b> | <b>0.294</b> | <b>0.272</b> |
|                                                                               |             | taud         | <b>2.050</b> | <b>2.030</b> | <b>2.058</b> | <b>2.200</b> | <b>2.154</b> | <b>2.103</b> | <b>2.183</b> | <b>2.331</b> | <b>2.421</b> | <b>2.375</b> | <b>2.169</b> | <b>2.032</b> |
| (42) <b>Clear-Sky Solar Irradiance</b>                                        | Ebn at Noon |              | <b>654</b>   | <b>764</b>   | <b>834</b>   | <b>853</b>   | <b>831</b>   | <b>795</b>   | <b>796</b>   | <b>827</b>   | <b>829</b>   | <b>763</b>   | <b>631</b>   | <b>561</b>   |
|                                                                               |             | Edh at Noon  | <b>71</b>    | <b>103</b>   | <b>126</b>   | <b>126</b>   | <b>140</b>   | <b>149</b>   | <b>136</b>   | <b>110</b>   | <b>88</b>    | <b>73</b>    | <b>64</b>    | <b>59</b>    |
|                                                                               | RadAvg      |              | <b>1.10</b>  | <b>2.18</b>  | <b>3.75</b>  | <b>4.93</b>  | <b>5.39</b>  | <b>5.80</b>  | <b>5.33</b>  | <b>4.59</b>  | <b>3.20</b>  | <b>1.94</b>  | <b>1.06</b>  | <b>0.76</b>  |
| (45) <b>All-Sky Solar Radiation</b>                                           | RadStd      |              | <b>0.06</b>  | <b>0.08</b>  | <b>0.15</b>  | <b>0.35</b>  | <b>0.40</b>  | <b>0.35</b>  | <b>0.46</b>  | <b>0.21</b>  | <b>0.27</b>  | <b>0.19</b>  | <b>0.07</b>  | <b>0.06</b>  |

**Historical Trends**

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | -1.84  | N/A     | N/A   | -0.52 | N/A         | N/A     | +44     | N/A     |
| Regional (10 neighbors) | N/A   | N/A     | -2.23  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |